DAY-1

Common Network IssuesTheory :Common hardware and software issues.Practical :Identify faulty devices in a simulated network.Task: Create a troubleshooting checklist.

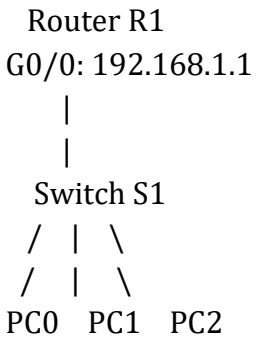
1. List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.
ANS :

No.	Hardware Issue	Description	Common Symptom
1	Damaged Ethernet Cable	The LAN cable may be broken, loose, or incorrectly connected.	Network icon shows disconnected or the switch port light remains OFF.
2	Faulty Network Interface Card (NIC)	The computer's Ethernet or Wi-Fi adapter may not be working correctly.	The network adapter is missing or shows an error in Device Manager.
3	Faulty Switch Port	A particular port on the network switch may stop functioning.	A device does not connect on one switch port but works when connected to another port.
4	Router Failure	The router may be powered off, damaged, or malfunctioning.	Multiple users lose access to other networks or the Internet at the same time.
5	Power Supply Problem	A router, switch, modem, or access point may not receive proper power.	No power/status LEDs are visible on the networking device.

Conclusion
Hardware problems can often be identified by checking cables, power, LEDs, network adapters, and switch/router ports before changing software settings.

2. Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify
which device is faulty and document the steps you took.
ANS :

Network Topology



IP Addressing

Device	IP Address	Subnet Mask	Default Gateway
Router R1 G0/0	192.168.1.1	255.255.255.0	—
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.20	255.255.255.0	192.168.1.1
PC2	192.168.1.30	255.255.255.0	192.168.1.1

Step 1 – Add Devices

Open **Cisco Packet Tracer**.
Add:

- 3 × PCs
- 1 × Cisco 2960 Switch
- 1 × Router

Arrange them as shown in the topology.

Step 2 – Connect the Devices

Use **Copper Straight-Through cables**.
Connect:
PC0 FastEthernet0 → Switch Fa0/1
PC1 FastEthernet0 → Switch Fa0/2
PC2 FastEthernet0 → Switch Fa0/3
Router G0/0 → Switch Fa0/24

Step 3 – Configure the Router

Open the router and go to:

CLI

Enter:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

Check the interface:

```
show ip interface brief
```

Expected result:

```
GigabitEthernet0/0    192.168.1.1    up    up
```

Step 4 – Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

```
IP Address:      192.168.1.10
Subnet Mask:     255.255.255.0
Default Gateway:  192.168.1.1
```

Step 5 – Configure PC1

```
IP Address:      192.168.1.20
Subnet Mask:     255.255.255.0
Default Gateway:  192.168.1.1
```

Step 6 – Configure PC2

```
IP Address:      192.168.1.30
Subnet Mask:     255.255.255.0
Default Gateway:  192.168.1.1
```

Step 7 – Test the Network Before Creating the Fault

Open:

PC0 → Desktop → Command Prompt

Run : ping 192.168.1.20

Then : ping 192.168.1.30

Then test the router : ping 192.168.1.1

If the network is configured correctly, the replies should be successful.

Example : Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Step 8 – Intentionally Create a Fault

For this practical, disconnect **PC2** from the switch.
Remove the cable:
PC2 FastEthernet0 → Switch Fa0/3
Now PC2 becomes the faulty/disconnected device.

Step 9 – Identify the Fault

From PC0, run : ping 192.168.1.20
Result : Successful
Then : ping 192.168.1.30
Result may show:
Request timed out.
or
Destination host unreachable.
This indicates that PC2 cannot be reached.

Step 10 – Check Physical Connection

Look at the topology.
PC0 and PC1 should have active green connection indicators.
PC2 will have no active link because its cable is disconnected.
Therefore, the problem is likely related to:
PC2
or
PC2's network connection

Step 11 – Use Packet Tracer Simple PDU

Select the **Add Simple PDU** tool.
Send a packet:
PC0 → PC1
It should show:
Successful
Now send:
PC0 → PC2
It should show:
Failed
This confirms that the problem is associated with PC2.

Step 12 – Repair the Fault

Reconnect:
PC2 FastEthernet0 → Switch Fa0/3
Wait until the link indicators turn green.
Run again:
ping 192.168.1.30
The ping should now succeed.

Result

The faulty device/network connection was identified as **PC2 because its Ethernet connection to the switch had been disconnected**.

Troubleshooting Steps Used

1. Checked physical connections.
2. Checked router and switch link indicators.
3. Used the ping command.
4. Compared communication with working PCs.
5. Used Packet Tracer's Simple PDU tool.
6. Located the disconnected PC.
7. Reconnected the Ethernet cable.
8. Retested connectivity.

3. Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks. Hint : Think about checking cables, router status lights, device settings, and running ping or speed tests.

ANS :

Slow Internet Troubleshooting Checklist

1. Check Router and Modem Power

Check whether the router and modem are powered ON.

Verify that the important status lights such as:

Power
Internet
Wi-Fi
LAN

are working normally.

2. Check Network Cables

Check the Ethernet cables connecting:

ISP/Modem → Router
Router → Computer

Make sure that cables are not loose or damaged.

3. Restart Router and Modem

Restart the modem and Wi-Fi router.

A restart can solve temporary problems such as:

- Router overload
- IP allocation problems
- Temporary software errors
- Connection instability

4. Check Wi-Fi Signal Strength

Check the Wi-Fi signal on the laptop or mobile device.

If the device is far from the router, move closer and test again.

Walls, furniture, and electronic devices can reduce Wi-Fi signal strength.

5. Check Number of Connected Devices

Too many devices connected at the same time can reduce Internet speed.

For example:

Smartphones
Laptops
Smart TVs
Gaming consoles
CCTV cameras

Disconnect unnecessary devices and test the connection again.

6. Run a Speed Test

Run an Internet speed test.

Compare the result with the speed provided by the ISP.

For example, if the Internet package is:

100 Mbps

but the device receives:

5 Mbps

there may be a Wi-Fi, router, ISP, or configuration issue.

7. Run a Ping Test

On Windows, open Command Prompt.

Test the router : ping 192.168.1.1

Then test Internet connectivity : ping 8.8.8.8

If the router ping is good but the Internet ping is very slow or fails, the problem may be with the ISP or WAN connection.

8. Check IP Configuration

Run:

ipconfig

Verify that the device has a valid:

IP Address

Subnet Mask

Default Gateway

DNS Server

An example valid configuration could be:

IPv4 Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

9. Check DNS

Sometimes the Internet works but websites load slowly because of DNS problems.

Test : ping 8.8.8.8

Then : ping google.com

If 8.8.8.8 works but google.com does not, the problem may be related to DNS.

10. Check Background Applications

Check whether applications are consuming Internet bandwidth.

Examples include:

- Windows Update
- Cloud backup
- Google Drive synchronization

- OneDrive synchronization
- Game downloads
- Video streaming

Open:

Task Manager → Performance / Processes

and check network usage.

11. Update Wi-Fi Drivers

Open:

Device Manager → Network Adapters

Check the wireless network adapter.

Update the driver if necessary.

An outdated or corrupted driver can cause slow or unstable Wi-Fi.

12. Check Router Placement and Interference

Place the router in an open and central location.

Avoid placing it near:

- Microwave ovens
- Bluetooth-heavy equipment
- Thick concrete walls
- Metal objects
- Other wireless routers

These can cause wireless interference.

13. Test Another Device

Connect another laptop or phone to the same Wi-Fi.

If only one device is slow, the problem is probably with that device.

If all devices are slow, the problem may be with:

Router

Modem

Wi-Fi interference

ISP connection

14. Contact the ISP

If all hardware and software checks are correct but the Internet remains slow, contact the Internet Service Provider.

Ask them to check:

Line quality

Network outage

Signal level

Bandwidth problems

Final Troubleshooting Flow

Slow Internet
|
v
Check Router/Modem Power
|
v
Check Cables
|
v
Restart Router
|
v
Check Wi-Fi Signal
|
v
Run Speed Test
|
v
Run Ping Test
|
v
Check IP/DNS Settings
|
v
Test Another Device
|
v
Check ISP

4. You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

ANS:

A user reports:

Instagram → Not working
YouTube → Not working
WhatsApp Web → Working

Because WhatsApp Web works, basic Internet connectivity is probably available. The problem may therefore involve DNS, browser settings, filtering, firewall rules, proxy configuration, or another network configuration issue.

First 5 Troubleshooting Steps

Step 1 – Confirm General Internet Connectivity

First verify whether the laptop is connected to the network.
Open Command Prompt and run : ping 8.8.8.8
If replies are received, basic Internet connectivity is working.
Example : Reply from 8.8.8.8: bytes=32 time=25ms TTL=117
This also suggests that the Wi-Fi adapter or Ethernet connection is functioning.

Step 2 – Test DNS Resolution

Run : ping youtube.com
And : nslookup youtube.com
Also test : nslookup instagram.com
If IP connectivity works but domain names cannot be resolved, the problem may be related to DNS.
Check the configured DNS using : ipconfig /all

Step 3 – Test Another Browser

Open YouTube and Instagram using another browser.
For example:
Chrome → Edge
or:
Edge → Chrome
If the websites work in another browser, the problem is probably browser-related.
Possible causes include:

- Browser extensions
- Corrupted cache
- Cookies
- Browser proxy settings
- Security extensions

Step 4 – Check Proxy, VPN, Firewall, and Security Software

Check whether a VPN or proxy is enabled.

In Windows check : **Settings → Network & Internet → Proxy**

Disable an unnecessary proxy temporarily for testing.

Also check :

- Windows Defender Firewall
- Antivirus
- Web filtering software
- VPN software

Some security software or network policies may block websites such as Instagram or YouTube while allowing WhatsApp Web.

Step 5 – Check Network Restrictions and DNS Filtering

Try the same websites on another device connected to the same Wi-Fi.

For example:

Laptop: Instagram → Not working

Phone on same Wi-Fi: Instagram → Not working

If both devices cannot access Instagram or YouTube, the restriction may be coming from:

- Router configuration
- DNS filtering
- Firewall
- Parental controls
- Network administrator policy
- ISP/network filtering

If Instagram and YouTube work on the phone but not on the laptop, the issue is more likely related to the laptop's:

- Browser
- DNS settings
- Firewall
- Proxy
- VPN
- Hosts file
- Security software

Final Troubleshooting Summary

When diagnosing network problems, troubleshooting should normally proceed from the lowest and simplest layer toward software and services:

1. Power

↓

2. Physical cables
   ↓
3. Network adapter
   ↓
4. Switch/Router connectivity
   ↓
5. IP configuration
   ↓
6. Ping test
   ↓
7. DNS
   ↓
8. Browser/Software
   ↓
9. Firewall/Proxy/VPN
   ↓
10. ISP or Network Administrator

This systematic approach helps identify whether the issue is caused by **hardware, software, or network configuration** without making unnecessary changes.